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TASK 3

UniProt

BLASTAlignPeptide searchID mappingSPARQL

Release 2026_02 | Statistics

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insulin

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Examples: Insulin, APP, Human, P05067, organism_id:9606

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Status

Reviewed (Swiss-Prot)
(5,400)

Unreviewed (TrEMBL)
(255,835)

Popular organisms

Human (1,887)

Mouse (1,792)

Rat (1,602)

Bovine (1,061)

Zebrafish (866)

Tools

Download (261k)

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Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☐ P01308	INS_HUMAN	Insulin[...]	INS	Homo sapiens (Human)	110 AA
☐ P06213	INSR_HUMAN	Insulin receptor[...]	INSR	Homo sapiens (Human)	1,382 AA
☐ P14735	IDE_HUMAN	Insulin-degrading enzyme[...]	IDE	Homo sapiens (Human)	1,019 AA
☐ P01317	INS_BOVIN	Insulin[...]	INS	Bos taurus (Bovine)	105 AA
☐ P01321	INS_CANLF	Insulin[...]	INS	Canis lupus familiaris (Dog) (Canis familiaris)	110 AA
☐ P01329	INS_CAVPO	Insulin[...]	INS	Cavia porcellus (Guinea pig)	110 AA

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>sp|P01308|INS_HUMAN Insulin OS=Homo sapiens OX=9606 GN=INS PE=1 SV=1
MALWMRLLPLLALLLWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAED
LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSI CSLYQLENYCN



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Start a New Modelling Project

Target Sequence(s):

(Format must be FASTA, Clustal, plain string, or a valid UniProtKB AC)

Target

Add Hetero Target

Reset

Project Title:

Untitled Project

Email:

Optional

Search For Templates

Build Model

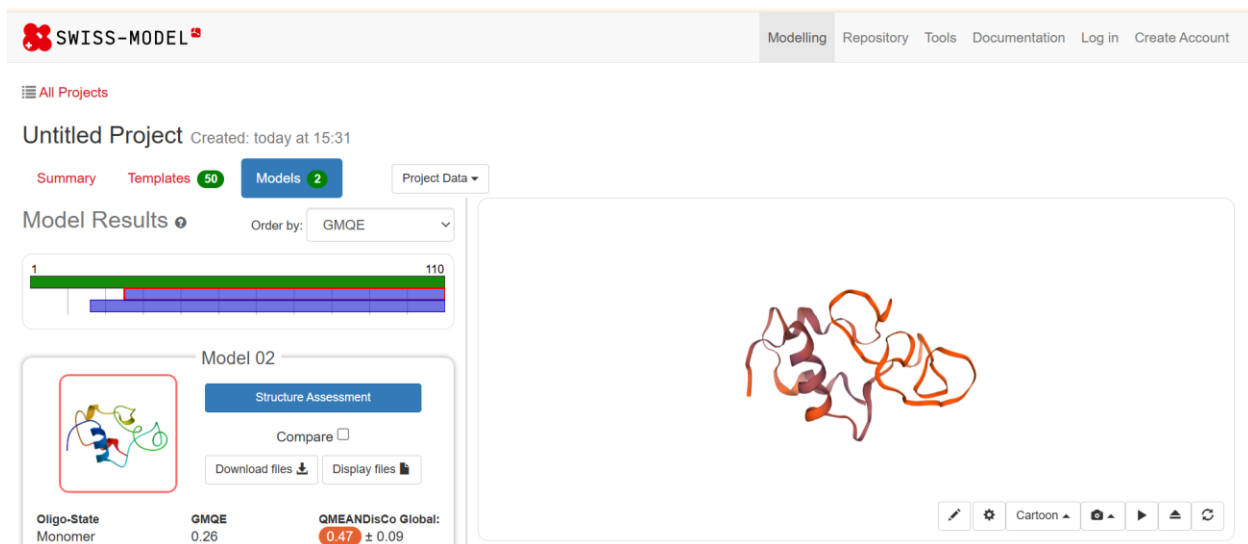
By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).

You are currently not logged in - to take advantage of the workspace, please [log in](#) or [create an account](#).

(There is no requirement to create an account to use any part of SWISS-MODEL, however you will gain the benefit of seeing a list of your previous modelling projects here.)

Supported Inputs

- Sequence(s) 
- Target-Template Alignment 
- User Template 
- DeepView Project 



Structural Features Report

Protein Structure Prediction Using SWISS-MODEL

A three-dimensional (3D) model of the target protein was generated using SWISS-MODEL through homology modeling. The predicted structure corresponds to a **monomeric protein** consisting of approximately **110 amino acid residues**.

Structural Characteristics

- **Model Type:** Homology-based 3D structure prediction
- **Oligomeric State:** Monomer
- **Protein Length:** 110 amino acids
- **Secondary Structure Elements:**
 - Multiple **α -helices** are visible in the central region.
 - Several **loop/coil regions** connect the structured domains.
 - No extensive β -sheet regions are apparent in the displayed model.
- **Overall Fold:**
 - The protein adopts a compact globular conformation.
 - Helical segments form the structural core, while flexible loops are located mainly at the terminal regions.

Model Quality Assessment

Parameter	Value
GMQE	0.26
QMEANDisCo Global	0.47 $\pm$ 0.09

Interpretation:

- The **GMQE score (0.26)** indicates moderate confidence in the predicted model.
- The **QMEANDisCo score (0.47)** suggests acceptable but not highly reliable structural quality.
- Conserved regions aligned with the template are predicted more confidently than flexible loop regions.

Biological Significance

The presence of conserved helical domains suggests that the protein may possess a stable structural core important for maintaining its biological function. Flexible loop regions may participate in molecular recognition, ligand binding, or protein-protein interactions.

Conclusion

The SWISS-MODEL analysis predicted a monomeric protein structure composed predominantly of α -helices connected by loop regions. The model exhibits a compact globular architecture with moderate confidence scores (GMQE = 0.26; QMEANDisCo = 0.47), indicating that the predicted structure is suitable for preliminary structural and functional analyses.